

Requirements Specification

Calculator App

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Abstract

The Calculator App is designed to solve basic linear equations. It allows users to enter an equation and displays the value of the unknown variable. The system provides a simple and quick way to solve equations and handles invalid input with an error message.

1 Introduction

The Calculator App is designed to solve basic linear equations. It allows users to enter an equation and displays the value of the unknown variable.

2 Purpose

The purpose of the application is to provide a simple and quick method for solving basic linear equations.

3 Scope

The system supports linear equations containing one unknown variable. It accepts an equation, processes it, and displays the solution.

4 Basic Functionality

The application will:

- Accept a linear equation from the user.

- Identify the unknown variable.
- Calculate its value.
- Display the result.
- Show an error for invalid input.

5 Requirements

The system shall allow users to enter and solve basic linear equations. It shall process the equation and display the calculated value of the unknown variable.

The system shall also detect invalid or unsupported equations and display an error message.

6 Examples

The application should solve equations such as:

- $2x + 5 = 15$
- $3x - 4 = 11$
- $5x = 25$
- $x + 7 = 12$

7 User Requirements

The user should be able to enter an equation easily and view the calculated answer clearly.

8 System Requirements

8.1 Hardware Requirements

- Computer or suitable device
- Display screen

8.2 Software Requirements

- Supported operating system
- Required software environment

9 Error Handling

The system should display an error when the entered equation is invalid or unsupported.

10 Conclusion

The Calculator App provides a simple way to solve basic linear equations. This document describes its main functionality and initial requirements.